

CLDV7111

ICE task1

Question1

1. A retail website automatically adds five more virtual servers to handle a sudden surge in traffic during a holiday sale and removes them once the traffic drops. Which NIST cloud characteristic does this illustrate?

- A. Measured service
- B. Rapid elasticity
- C. On-demand self-service
- D. Resource pooling

2. In the ASP.NET Core MVC architecture, which component is primarily responsible for handling user input, interacting with the data model, and choosing which view to display?

- A. Middleware
- B. Model
- C. View
- D. Controller

3. When comparing Azure App Service to managing your own Virtual Machine (IaaS), what is a significant advantage of using the App Service?

- A. Lower overall latency for local users
- B. Managed patching and infrastructure updates
- C. Full access to the Operating System kernel
- D. Requirement to manually install web server software

4. During the deployment of the EventEase system, you must connect your Web App to an Azure SQL Database. What is the name of the string that contains the server address, database name, and credentials?

- A. Deployment Slot
- B. App Service Plan
- C. Connection String
- D. Resource Group ID

5. What is the primary role of the Azure Container Registry (ACR) in a cloud development workflow?

- A. To store and manage private Docker container images
- B. To host and run live websites
- C. To monitor the CPU usage of running containers
- D. To manage SQL database schemas

6. In Azure Container Registry, how can a developer distinguish between different versions (e.g., 'v1', 'v2', 'latest') of the same container image?

- A. By applying specific Tags to the image
- B. By changing the name of the Resource Group
- C. By creating a new Subscription for each release
- D. By using different Registry URLs for each version

7. Which Azure service allows you to run a single container quickly without having to provision or manage any underlying Virtual Machines?

- A. Azure Container Instances (ACI)
- B. Azure Virtual Machines
- C. Azure Static Web Apps
- D. Azure DevOps

8. What is a major security benefit provided by Azure Container Instances (ACI) when running applications?

- A. Requirement for all users to have Root access
- B. Hypervisor-level isolation for each container group
- C. Automatic physical shredding of hard drives
- D. Exclusively using public IP addresses

9. Which cloud characteristic allows a service provider to serve multiple customers using a shared infrastructure while keeping their data separate?

- A. Resource pooling
- B. On-demand self-service
- C. Broad network access
- D. Measured service

10. If you are deploying a long-running web application that requires automatic horizontal scaling and 'deployment slots' for blue/green testing, which service should you choose?

- A. Azure Container Registry
- B. Azure SQL Database
- C. Azure Container Instances (ACI)
- D. Azure App Service

(20)

Question 2: True or False

1. Resource Pooling' in cloud computing means that a single user has exclusive access to a specific piece of physical hardware, such as a dedicated hard drive.
2. In the ASP.NET Core MVC pattern, the 'Controller' acts as the intermediary that handles user requests and updates the Model.
3. Azure App Service is classified as Infrastructure as a Service (IaaS) because you must manually manage the operating system patching.

4. The Azure Container Registry (ACR) is used primarily for storing and managing container images, rather than executing the containers themselves.
5. ASP.NET Core MVC is a cross-platform framework, meaning it can run on Windows, Linux, and macOS.

(5)

Question 3

Draw the:

1. MVC Model
2. ERD between a **Playlist** and a **Song**

(10)

Total 35